

Pravallika Thota *Software Engineer*

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Summary

AI/ML Developer with 2+ years of experience building intelligent applications using Machine Learning, Generative AI, and Large Language Models (LLMs). Experienced in developing AI agents, Retrieval-Augmented Generation (RAG) systems, NLP solutions, and end-to-end AI applications. Skilled in model development, fine-tuning, prompt engineering, API development, and cloud-based AI integrations. Proficient in Python, modern AI frameworks, FastAPI, and full-stack web technologies, with a strong focus on designing scalable, production-ready AI solutions and automating business workflows.

Professional Experience

AI/ML Developer

TSSS INFOTECH & INFRA PVT LTD

12/2024

Hyderabad

Education

BSC

YSRH University

08/2020 – 06/2024

Tadepalligudem

Skills

Python	LangChain
LangGraph	RAG
MCP	NLP
Fine Tuning	Fast Api
Machine Learning	Neural Networks
Transformers	Google ADK
Crew AI	Agno
Pytorch	ScikitLearn
Pandas	NumPy
Amazon Connect	Amazon Lex
Supabase	Hugging Face
HTML	CSS
JavaScript	React

Projects

Project 1: DealWallet – AI-Powered E-Commerce Platform

Role: AI/ML Developer

Technologies: Python, PyTorch, Scikit-learn, Pandas, NumPy, LangChain, LangGraph, FastAPI, Hugging Face, NLP, NLTK, spaCy, RAG, MCP, Twilio, Amazon Connect, Amazon Lex, React, JavaScript, HTML, CSS, Supabase, ClickHouse, Meilisearch, OpenCV, Tesseract OCR, GitHub

- Developed an AI-powered chatbot using LangChain and LangGraph with Agentic RAG to retrieve contextual information from enterprise databases.
- Built multi-agent workflows with agent-to-agent communication and Human-in-the-Loop (HITL) capabilities for complex business processes.
- Implemented a voice assistant using Amazon Connect, Amazon Lex, LangGraph, and Twilio to enable intelligent voice interactions.
- Integrated WhatsApp, Telegram, SMS, and voice call services using Twilio for omnichannel customer engagement.
- Developed machine learning models for customer churn prediction, payment fraud detection, and product price trend forecasting using PyTorch and Scikit-learn.
- Built NLP pipelines for customer sentiment analysis using NLTK and spaCy to derive actionable insights from user feedback.
- Developed OCR-based document processing solutions using OpenCV and Tesseract for automated data extraction.
- Designed scalable REST APIs using FastAPI to expose AI/ML services and integrate them with frontend and backend applications.
- Leveraged ClickHouse, Meilisearch, and Supabase for efficient data storage, semantic search, and retrieval to support AI-driven applications.
- Collaborated with frontend components built using React, JavaScript, HTML, and CSS while following GitHub-based version control and development workflows.

Project 2: AI-Powered VPN Customer Support Chatbot

Role: AI/ML Developer

Technologies: Python, FastAPI, NLP, Machine Learning, Fine-Tuned LLMs, LangChain, RAG, Hugging Face, REST APIs

- Developed an AI-powered customer support chatbot to automate VPN troubleshooting and enhance customer experience using NLP and fine-tuned LLMs.
- Designed a multi-level intent classification system to accurately identify customer issues, including authentication, connectivity, billing, subscription, and account management requests.
- Fine-tuned language models and implemented contextual intent recognition to improve response accuracy and deliver personalized, context-aware support.
- Built intelligent troubleshooting workflows integrated with backend knowledge repositories using RAG, enabling automated issue diagnosis and resolution.
- Developed scalable REST APIs using FastAPI to integrate chatbot services with enterprise support platforms.
- Automated high-volume customer support interactions, reducing manual support effort, improving response times, and increasing self-service efficiency.

Project 3: ATS-Friendly Resume Analyzer

Role: AI/ML Developer

Technologies: Python, NLP, spaCy, NLTK, Machine Learning, FastAPI, OCR, OpenCV, Tesseract, React, REST APIs

- Developed an AI-powered resume analyzer to evaluate resumes against job descriptions using NLP and machine learning techniques.

- Implemented resume parsing, keyword extraction, skill matching, and ATS compatibility scoring to identify gaps and improve resume quality.
- Built OCR-based document processing using OpenCV and Tesseract to extract information from PDF and image-based resumes.
- Developed scalable REST APIs with FastAPI and integrated the solution with a React-based user interface for real-time resume analysis and recommendations.

Project 4: AI-Powered Healthcare Report Processing & Virtual Care Platform

Role: AI/ML Developer

Technologies: Python, Agno, MCP, LLMs (Gemma), Multi-Agent AI, FastAPI, RAG, Twilio

- Developed an AI-powered healthcare platform to automate medical report processing, patient engagement, and virtual healthcare services.
- Built a multi-agent workflow using Agno and MCP to extract structured data from health reports using Gemma LLMs and streamline report processing.
- Implemented a Human-in-the-Loop (HITL) validation workflow where extracted values were verified before storing report data in the database and delivering finalized reports to patients.
- Developed an AI chatbot enabling patients to interact with doctors, receive assistance, and book appointments through conversational AI.
- Implemented a voice-based AI assistant to facilitate patient-doctor interactions and appointment scheduling via phone calls.
- Designed scalable backend APIs to integrate AI services, automate healthcare workflows, and improve operational efficiency.